Stat 266 – Unit 7 – Video Set 1 and 2 – Class Examples

Summary:

Distributions Related to the *Normal* Distribution: $Y_1,\ldots,Y_n \overset{ind.}{\sim} Normal(\mu_Y,\sigma_Y^2)$.

- $\bar{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i \sim Normal(\mu_{\bar{Y}} = \mu_Y,\ \sigma_{\bar{Y}}^2 = \sigma_Y^2/n)$.
- $\sum_{i=1}^{n} Z_i^2 = \sum_{i=1}^{n}\left(\frac{Y_i - \mu_Y}{\sigma_Y}\right)^2 \sim \chi_n$.
- $\frac{(n-1)S^2}{\sigma_Y^2} = \frac{\sum_{i=1}^{n}(Y_i - \bar{Y})^2}{\sigma_Y^2} \sim \chi_{n-1}^2$.

- Consider independent random variables Z and W where
  a) $Z \sim N(0,1)$,
  b) $W \sim \chi_v^2$.

  $\Rightarrow T = \frac{Z}{\sqrt{W/v}} \sim t_v$ (A t distribution with v degrees of freedom.)
- Consider independent random variables W_1 and W_2 where
  a) $W_1 \sim \chi_{v_1}^2$,
  b) $W_2 \sim \chi_{v_2}^2$.

  $\Rightarrow F = \frac{W_1/v_1}{W_2/v_2} \sim F_{v_2}^{v_1}$ (An F distribution with v_1 and v_2 degrees of freedom.)

- **Central Limit Theorem:** $\frac{\bar{Y} - \mu_Y}{\sigma_Y/\sqrt{n}} \to Normal(\mu = 0,\ \sigma^2 = 1)$ as $n \to \infty$.

Review:

Proposition: The expectation is a linear function of random variables.

i.e. $E(aX+bY) = a\cdot E(X) + bE(Y)$. ①

The variance $V(X) = E((X-EX)^2)$. So by formula ①, we have $E(X^2-(2EX)\cdot X+(EX)^2) = E(X^2)-(EX)^2$.

claim $V(aX) = a^2V(X)$.

$V(aX) = E((aX)^2) - (E(aX))^2 = E(a^2X^2) - (aE(X))^2$

$= a^2EX^2 - a^2(EX)^2 = a^2V(X)$.

Thm 1: Let $Y_1, \cdots, Y_n$ be i.i.d samples from normal distribution $N(\mu, \sigma^2)$

Then $\bar{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i$ is of Normal distribution $N(\mu, \frac{\sigma^2}{n})$.

since $Y_i \sim N(\mu, \sigma^2)$. and Y_i are i.i.d $\sum_{i=1}^{n} Y_i$ is also Normal distribution.

$$E(\bar{Y}) = E\left(\frac{1}{n}\sum_{i=1}^{n} Y_i\right) = \frac{1}{n}\sum_{i=1}^{n} E(Y_i) = \frac{1}{n}\cdot n\mu = \mu.$$

$$V(\bar{Y}) = V\left(\frac{1}{n}\sum_{i=1}^{n} Y_i\right) = \frac{1}{n^2}\sum_{i=1}^{n} V(Y_i) = \frac{1}{n^2}\cdot n\sigma^2 = \frac{\sigma^2}{n}.$$

Thm 2: Let $Y_1, \cdots, Y_n$ are i.i.d samples of $N(\mu, \sigma^2)$. Then $\sum_{i=1}^{n}\left(\frac{Y_i-\mu}{\sigma}\right)^2$.

is of chi-square distribution with n degrees of freedom.

proof. $\frac{Y_i-\mu}{\sigma} \sim N(0, 1)$. standard Normal

and $\frac{Y_i-\mu}{\sigma}$ are independent since $Y_1, \cdots, Y_n$ are independent.

Then $\sum_{i=1}^{n} \left(\frac{Y_i-\mu}{\sigma}\right)^2 \sim$ chi-square distribution with n freedom.

Thm 3: Let $Y_1, \dots, Y_n$ be i.i.d samples of $N(\mu, \sigma^2)$.

$\frac{(n-1)S^2}{\sigma^2} = \frac{1}{\sigma^2} \sum_{i=1}^{n} (Y_i - \bar{Y})^2$ has chi square distribution with $(n-1)$ degree of freedom.

And S and $\bar{Y}$ are independent.

$\sum_{i=1}^{n} \left(\frac{Y_i-\mu}{\sigma}\right)^2$ is χ^2-distribution of freedom n.

$$\|$$

$$\sum_{i=1}^{n} \left(\frac{Y_i - \bar{Y} + \bar{Y} - \mu}{\sigma}\right)^2 = \sum_{i=1}^{n} \frac{(Y_i - \bar{Y})^2}{\sigma^2} + \sum_{i=1}^{n} \frac{2(Y_i - \bar{Y})(\bar{Y} - \mu)}{\sigma^2}$$

$$+ \sum_{i=1}^{n} \left(\frac{\bar{Y} - \mu}{\sigma}\right)^2 = \frac{(n-1)S^2}{\sigma^2} + \frac{2(\bar{Y} - \mu)}{\sigma^2} \underbrace{\sum_{i=1}^{n} (Y_i - \bar{Y})}_{=0} + \sum_{i=1}^{n} \left(\frac{\bar{Y} - \mu}{\sigma}\right)^2$$

$$= \frac{(n-1)S^2}{\sigma^2} + \sum_{i=1}^{n} \left(\frac{\bar{Y}-\mu}{\sigma}\right)^2. \sim \chi^2\text{- distribution of freedom } n-$$

Since $\sum_{i=1}^{n} \left(\frac{\bar{Y}-\mu}{\sigma}\right)^2 = n\left(\frac{\bar{Y}-\mu}{\sigma}\right)^2 = \left(\frac{\bar{Y}-\mu}{\sqrt{n}\,\sigma}\right)^2$

$\frac{\bar{Y}-\mu}{\sqrt{n}\,\sigma} \sim N(0,1)$. If we further assume S^2 and $\bar{Y}$ are independent,

then $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2$ distribution of freedom $n-1$.

Example A : Suppose that $Z_1, \ldots, Z_4$ is a random sample from a standard normal distribution. What is $V[U]$, where $U = (Z_1 + 2Z_2 + 2Z_3 + 4Z_4)^2$?

$Z_1 \sim N(0,1)$ $\quad 2Z_2 \sim N(0,4)$ $\quad 2Z_3 \sim N(0,4)$ $\quad 4Z_4 \sim N(0,16)$.

$Z_1 + 2Z_2 + 2Z_3 + 4Z_4 \sim N(0,25)$.

$$\frac{Z_1 + 2Z_2 + 2Z_3 + 4Z_4}{5} \sim N(0,1).$$

$$\left(\frac{Z_1 + 2Z_2 + 2Z_3 + 4Z_4}{5}\right)^2 \sim \chi_1^2$$

$$V\left(\frac{(Z_1 + 2Z_2 + 2Z_3 + 4Z_4)^2}{25}\right) = 1 + 1 = 2.$$

$$V\left((Z_1 + 2Z_2 + 2Z_3 + 4Z_4)^2\right) = 2 \times 25^2.$$

$$= 1250.$$

Example B : Suppose that $Z_1, \ldots, Z_n$ are independent standard normal random variables.

If $V = \dfrac{c(Z_1 + Z_2)}{\left(Z_3^2 + Z_4^2 + Z_5^2\right)^{1/2}} \sim t_{df}$, what are c and df ?

Note that if $X \sim N(0,1)$ $W \sim \chi^2_d$

$$\frac{X}{\sqrt{\frac{W}{d}}} \sim t_d \text{ distribution.}$$

since $Z_1 + Z_2 \sim N(0, 2)$. $\dfrac{Z_1+Z_2}{\sqrt{2}} \sim N(0,1)$.

$Z_3^2 + Z_4^2 + Z_5^2 \sim \chi^2_3$ distribution

So $$\frac{Z_1+Z_2}{\sqrt{2}} \Big/ \sqrt{\frac{Z_3^2+Z_4^2+Z_5^2}{3}} \sim t_3 \quad \frac{\frac{Z_1+Z_2}{\sqrt{2}}}{\frac{\sqrt{Z_3^2+Z_4^2+Z_5^2}}{\sqrt{3}}} = \underbrace{\frac{\sqrt{3}}{\sqrt{2}}}_{= c} \cdot \frac{Z_1+Z_2}{\sqrt{Z_3^2+Z_4^2+Z_5^2}}$$

Example C : Suppose that $Z_1, \ldots, Z_n$ are independent standard normal random variables.

If $V = c\dfrac{(Z_1 - Z_2)^2}{Z_3^2 + Z_4^2 + Z_5^2 + Z_6^2} \sim F_{v_2}^{v_1}$, what are c, v_1, and v_2?

$V(Z_1 - Z_2) = V(Z_1) + V(Z_2) = 2$ so $Z_1 - Z_2 \sim N(0, 2)$.

$\dfrac{Z_1 - Z_2}{\sqrt{2}} \sim N(0,1)$ $\quad \left(\dfrac{Z_1 - Z_2}{\sqrt{2}}\right)^2 \sim \chi_1^2$.

Now $Z_3^2 + Z_4^2 + Z_5^2 + Z_6^2 \sim \chi_4^2$.

So
$$\frac{\left(\frac{Z_1 - Z_2}{\sqrt{2}}\right)^2}{(Z_3^2 + Z_4^2 + Z_5^2 + Z_6^2)/4} \sim F_4^1.$$

$$\| $$

$$\frac{4(Z_1 - Z_2)^2}{2(Z_3^2 + Z_4^2 + Z_5^2 + Z_6^2)} = c \cdot \frac{(Z_1 - Z_2)^2}{(Z_3^2 + Z_4^2 + Z_5^2 + Z_6^2)} \Rightarrow c = 2.$$

Example D: Suppose the time it takes to serve a customer at particular coffee shop follows an Exponential distribution with mean 30 seconds. What is the probability that it will take less than 45 minutes to serve 100 customers?

Let X_i be the time to serve ith customer.

$$P\left(\sum_{i=1}^{100} X_i < 45 \text{ min}\right) = P\left(\bar{X} < \frac{45}{100} \text{ min}\right) = P(\bar{X} < 27 \text{ secs})$$

Now X_i are i.i.d samples of $X \sim$ Exponential

$$f_X(x) = \begin{cases} \frac{1}{30} e^{-\frac{x}{30}} & x > 0 \\ 0 & \text{otherwise} \end{cases} \qquad E(X) = 30 \qquad V(X) = 30^2.$$

$$Y = \sqrt{100}\,\frac{\bar{X} - 30}{30} \sim N(0,1).$$

$$P(\bar{X} < 27) = P\left(10\left(\frac{\bar{X} - 30}{30}\right) < \left(\frac{27 - 30}{30}\right) \times 10\right) = P(Y < -1).$$

Check the table of Normal distribution.

Example E: You are playing game. The random variable, ***Y***, represents your possible winnings. The probability distribution of Y is

Y	-3	-2	0	1	2
$P(\boldsymbol{Y} = y)$	0.25	0.25	0.10	0.10	0.30

What is the probability that you are up money after 100 plays?

Let Y_i be the number of your winnings. $0 \le i \le 100$.

$$p\left(\sum_{i=1}^{n} Y_i > 0\right). = p\left(\frac{1}{n}\sum_{i=1}^{n} Y_i > 0\right). = p(\bar{Y} > 0).$$

$$E(Y) = (-3)\times 0.25 + (-2)\times 0.25 + 0\times 0.10 + 1\times 0.1 + 2\times 0.3$$

$$= -3\times 0.25 + 0.1 + 0.6 = -0.55.$$

$$V(Y) = 9\times 0.25 + 4\times 0.25 + 0\times 0.10 + 1\times 0.10 + 4\times 0.30 - (-0.55)^2 = 4.2475.$$

By CLT: $X \approx \sqrt{100}\left(\frac{\bar{Y}+0.55}{\sqrt{4.2475}}\right) \sim N(0,1)$.

$$P(\bar{Y}>0) = P\left(10\times\frac{\bar{Y}+0.55}{\sqrt{4.2475}} > \frac{0.55}{\sqrt{4.2475}}\times 10\right).$$

$$= P\left(X > \frac{0.55}{\sqrt{4.2475}}\times 10\right). = 1 - P\left(X < \frac{0.55}{\sqrt{4.2475}}\times 10\right).$$

Example F: Suppose the number of goals a team scores in a hockey game follows a Poisson distribution with mean 3. The team has an extremely consistent goalie that allows exactly 2 goals per game. Thus, the team needs at least 3 goals to win any given game. What is the probability they will get 100 wins in the next two seasons? (164 games in 2 seasons.)

Since X (the number of Goals in each game) $\sim$ Poisson distribution

with $E(X) = 3$, $P(X=k) = \dfrac{3^k e^{-3}}{k!}$ $(k=0,1,2\ldots)$.

Let Y be a random variable s.t. $Y = \begin{cases} 1 & \text{if } X \geq 3 \\ 0 & \text{if } X \leq 2 \end{cases}$

Then $P(Y=0) = P(X=0) + P(X=1) + P(X=2) = e^{-3} \cdot \frac{17}{2}$.

$P(Y=1) = 1 - e^{-3} \cdot \frac{17}{2}$. $E(Y) = 1 - e^{-3} \cdot \frac{17}{2} \approx 0.577$

$V(Y) = 0^2 (e^{-3} \cdot \frac{17}{2}) + 1^2 (1 - e^{-3} \cdot \frac{17}{2}) - (0.577)^2 = 0.577 - (0.577)^2$

$= 0.243671$

Then this team win at least 100 games out of 164 games.

That means we sample 164 Y_i from Y, and average $\bar{Y} \geq \frac{100}{164}$.

$$P(\text{team win more than } 100) = P\left(\bar{Y} > \frac{100}{164}\right)$$

$$= P\left(\sqrt{164}\,\frac{\bar{Y} - 0.577}{\sqrt{0.243671}} > \frac{\left(\frac{100}{164} - 0.577\right)\sqrt{164}}{\sqrt{0.243671}}\right)$$

compute this number & use table of Normal distribution.

Example G: A machine produces trick coins where P(heads) = 0.55. Half way during production someone changes the factory settings so that P(heads) = 0.50. At the end of the day the factory ended up with a bin of coins where half are fair coins and half are the trick coins. The company decides to flip each coin 1000 times. If the result is at least 525 heads they will judge the coin to be a trick coin. How many fair coins will be classified as trick coins? How many trick coins will be classified as fair?

Let's firstly compute the Probability that one fair coin is judged by tricky coin.

Let $X = \begin{cases} 1 & \text{a coin is head} \\ 0 & \text{a coin is tail.} \end{cases}$

If we have a fair coin $p(X=1) = \frac{1}{2}$ $p(X=0) = \frac{1}{2}$.

$E(X) = \frac{1}{2}$ $V(X) = \frac{1}{2} - (\frac{1}{2})^2 = \frac{1}{4}$.

Let $X_1, \cdots, X_{1000}$ be 1000 independent samples from X.

$$P\left(\sum_{i=1}^{1000} X_i > 0.525\right) = P\left(\bar{X} > 0.525\right) = P\left(\sqrt{1000}\,\frac{(X-\frac{1}{2})}{\sqrt{\frac{1}{4}}} \geq\right.$$

$$\left.\sqrt{1000} \times \frac{(0.525-0.5)}{\sqrt{\frac{1}{4}}}\right) = P\left(\sqrt{1000}\,\frac{(\bar{X}-\frac{1}{2})}{\sqrt{\frac{1}{4}}} \geq 0.05 \times \sqrt{1000}\right).$$

By CLT $\dfrac{\sqrt{1000}\,(\bar{X}-\frac{1}{2})}{\sqrt{\frac{1}{4}}} \sim N(0,1)$.

If we have tricky coin: $P(X=1) = 0.55$ $P(X=0) = 0.45$.

$E(X) = 0.55$ $V(X) = (1-0.55)^2 \times 0.55 + (0-0.55)^2 \times 0.45$

$= (0.45)^2 \times 0.55 + (0.55)^2 \times 0.45 = 0.55 \times 0.45 = 0.2475$

Let $X_1, \dots, X_{1000}$ be 1000 samples from X.

$$P\left(\sum_{i=1}^{1000} X_i < 525\right) = P\left(\bar{X} < 0.525\right) = P\left(\frac{\sqrt{1000}(\bar{X} - 0.55)}{\sqrt{0.2475}} < \frac{0.525 - 0.55}{\sqrt{0.2475}} \times \sqrt{1000}\right)$$

$$\frac{\sqrt{1000}(\bar{X} - 0.55)}{\sqrt{0.2475}} \sim N(0,1).$$

compute this and check the table.